

Experimental Theorems for Paper B’s F1 / Slack-Two Resonance Window

RS-MCA experimental notes

June 18, 2026

Abstract

This note records restricted experimental theorems and heuristics extracted from the Cycle 14–18 audits in the RS-MCA repository. The main new item is a divisibility gate for the non-coprime resonance branch in the quadratic-extension, slack-two, three-co-support toy window. The statements are aimed at Paper B’s residue-line normal form and per-fiber collision program, but they are not a proof of the corrected MCA conjecture and have no direct protocol consequence.

The note also records an attribution audit for a separate AI-generated packet whose item (d) overlapped substantially with Theorem 1.13 of Ben-Sasson, Carmon, Habock, Kopparty, and Saraf, *On proximity gaps for Reed–Solomon codes*, to appear at STOC 2026. That overlap should be treated as an imported external theorem/proof pattern, not as a new contribution of this repository.

1 Status and Scope

All results in this note live in the experimental folder and should be read as restricted algebraic facts or proof targets. We do not edit or supersede the main papers A–D.

The source map is:

- Paper A, `tex/RS_disproof_v3.tex`, refutes the old no-slack smooth-domain support-wise line-MCA statement.
- Paper B, `tex/slackMCA_v3.tex`, is the main target here. In particular, this note is a small local contribution to the residue-line normal form around `def:residue`, `thm:normalform`, `prob:perfiber`, and `conj:B`.
- Paper D, `tex/cs25_cap_v4.tex`, gives the universal field-size cap and imported list-to-agreement route. Nothing below changes that cap.
- Paper C, `tex/snarks_v4.tex`, records protocol ledgers. Nothing below may be consumed by a protocol without a separate conversion theorem.

The common restricted setting is

$$B = \mathbb{F}_p, \quad F = \mathbb{F}_{p^2}, \quad D = \mathbb{F}_p, \quad t = \sigma = 2, \quad j = 3,$$

and we work off the tangent/global endpoint

$$R_0 = \{ [W]_E \wedge [B_{\text{num}}]_E = 0 \}.$$

The field ledgers remain distinct:

$$q_{\text{gen}} = p, \quad q_{\text{line}} = p^2.$$

The regime is sub-reserve, since $\eta = \sigma/n = 2/n$. Thus none of the following proves a corrected-reserve list, CA, MCA, line-decoding, SNARK, or protocol theorem.

2 External Proximity-Gap Attribution Audit

Status: AUDIT / ATTRIBUTION. This section concerns a separate AI-generated packet with four advertised results, labeled here only as (a)–(d). It should guide later edits to the main papers, but it is not itself a new theorem.

The cited external source is:

Ben-Sasson–Carmon–Habock–Kopparty–Saraf, *On proximity gaps for Reed–Solomon codes*.

Swastik Kopparty’s paper page lists the same paper with coauthors Eli Ben-Sasson, Dan Carmon, Ulrich Habock, and Shubhangi Saraf. The paper is reported to appear at STOC 2026. The relevant result to audit against is Theorem 1.13, with the subgroup/sumset construction and proof pattern around Theorem 7.1.

Attribution correction

Any repository text or AI output claiming the proof of result (d) as a new contribution should be treated as incorrect. The proof strategy of (d) is, up to notation and presentation, essentially the proof of Theorem 1.13 in the external proximity-gap paper. In particular, even notation such as using ℓ for the size of the chosen subset appears to track the external paper closely enough that the safe status is:

result (d): imported theorem/proof pattern, not original.

Before any main-paper promotion, the text should cite the external theorem, state exactly which hypotheses are being imported, and explain whether the RS-MCA notation changes the object or merely renames it.

How to treat the four advertised results

Item	Current limitation	Main-paper handling
(a)	Works only at slack on the order of $1/\sqrt{n}$.	Use at most as a weak-slack comparison point. It does not clear Paper B’s corrected-reserve objective and should not be stated as evidence for capacity-minus-explicit-floors MCA.
(b)	Applies only to three Fermat-prime cases.	Treat as finite-prime evidence or a worked example. It is not a uniform smooth-domain theorem and should not be allowed to drive asymptotic claims.
(c)	Requires exponential field size.	Treat as a large-field construction or sanity check. It is not directly usable for the polynomial-field or Proximity Prize parameter ledger.
(d)	Most mathematically relevant, but the proof is essentially external.	Import, cite, and adapt from Theorem 1.13 of the proximity-gap paper. Do not present it as an AI/repository proof. Audit the exact object before using it in Paper B, D, or C.

Main-paper integration plan

Paper B. Add a related-work and dependency paragraph near the local-limit/frontier discussion saying that the proximity-gap theorem of Ben-Sasson–Carmon–Habock–Kopparty–Saraf is the source for the item-(d) proof pattern. If a Paper B statement uses the theorem, state it as an imported theorem with exact parameters: domain, rate, slack, field-size regime, and whether the object is a proximity gap, list statement, CA/MCA statement, or line-decoding statement. Do not let the $1/\sqrt{n}$ -slack result (a), the three-Fermat-prime result (b), or the exponential-field result (c) substitute for the corrected-reserve MCA conjecture.

Paper D. Keep the Crites–Stewart list-to-agreement import separate from this proximity-gap import. If item (d) is relevant to the universal-cap narrative, it should be cited as external background or as a separately imported theorem, not folded into the cap proof unless the field ledger and radius normalization are checked line by line.

Paper C. Protocol ledgers should not consume item (d) merely because it is a Reed–Solomon proximity theorem. A protocol-facing use still needs a precise conversion into the consumed object: base-list size, interleaved-list size, CA, MCA, line-decoding, or curve-MCA. Until such a conversion is proved, the ledger status is CONDITIONAL/IMPORTED at best.

Paper A. No change is forced. These results do not rescue the old no-slack smooth support-wise MCA theorem refuted by Paper A. They should be cited only if they clarify the landscape around proximity gaps and explicit slack regimes.

Notation hygiene

When this material is eventually moved into the main papers, avoid copying the external notation silently. If the external proof uses ℓ for a chosen subset size, either keep that notation with an explicit citation or rename it, for example to ℓ_{pg} , so it is not confused with RS-MCA slack t, σ ,

support size, list size, or line-decoding parameters. The ledger notation must still distinguish q_{gen} , q_{line} , q_{chal} , B , and F .

Checklist before promotion

- Add a BibTeX entry for the proximity-gap paper before citing it in Papers A–D.
- Replace any claim of “new proof of (d)” by “adapted from Theorem 1.13 of Ben-Sasson–Carmon–Habock–Kopparty–Saraf.”
- Verify whether item (d)’s theorem object is list/proximity, CA, MCA, line-decoding, or something only convertible under an additional theorem.
- Record which of (a)–(d) operate at $1/\sqrt{n}$ slack, three Fermat primes, exponential field size, and imported-theorem status.
- Re-run the field ledger: do not let an extension field pay a generated-field entropy bill unless the cited theorem explicitly does so.

3 Notation

The base/generated field is $B = \mathbb{F}_p$. The line field is the quadratic extension $F = \mathbb{F}_{p^2} = B(\alpha)$, with basis $\{1, \alpha\}$ over B and $\alpha^2 = \nu \in B$. The domain is $D = \mathbb{F}_p$, so $n = |D| = p$.

A degree-two residue-line datum consists of:

$$E \in F[X], \quad \deg E = 2, \quad E(x) \neq 0 \text{ for all } x \in D,$$

and a numerator

$$B_{\text{num}} \in F[X], \quad \deg B_{\text{num}} < 2.$$

We write

$$A = F[X]/E, \quad [P]_E \in A, \quad b = [B_{\text{num}}]_E.$$

The anchor word on D is interpolated by a polynomial W . The operation $u \wedge v$ means the determinant of the coordinate vectors of $u, v \in A$ in a chosen F -basis of the two-dimensional F -space A . Off R_0 , the pair $\{[W]_E, b\}$ is such a basis.

The co-support has size $j = 3$. If $T \subset D$ is the three-point co-support, define

$$\tau_1 = e_1(T), \quad \tau_2 = e_2(T), \quad \tau_3 = e_3(T).$$

All valid co-supports have $\tau_i \in B$. The quantity C_2 denotes the number of distinct slopes produced by the relevant landing branch after restricting to valid split triples $T \subset \mathbb{F}_p$. Bounding landing points therefore bounds C_2 , but a sharp slope theorem may require more because many landing points can share a slope.

4 Cycle 14 Normal Form

The Cycle 14 audit reduces the $t = 2, j = 3$ landing equation to two affine forms in the elementary co-support parameters

$$\tau = (\tau_1, \tau_2, \tau_3).$$

Lemma 1 (Affine wedge normal form). *Assume $\{[W]_E, b\}$ is an F -basis of A . Then the landing equation can be written in the form*

$$\iota(\tau) = A_0(\tau_1, \tau_2) - \tau_3[W]_E, \quad \mu(\tau) = B_0(\tau_1, \tau_2) - \tau_3 b,$$

where

$$A_0 = p_1[W]_E + p_2 b, \quad B_0 = q_1[W]_E + q_2 b,$$

and $p_i, q_i \in F$ are affine-linear functions of (τ_1, τ_2) . If the basis is normalized so $[W]_E \wedge b = 1$, then

$$\Delta = \iota \wedge \mu = (p_1 - \tau_3)(q_2 - \tau_3) - p_2 q_1.$$

Proof. The Cycle 14 quotient calculation, which is a concrete instance of Paper B's residue-line normal form, gives the displayed affine forms. Expanding in the ordered basis $\{[W]_E, b\}$, the coordinate vector of ι is $(p_1 - \tau_3, p_2)$, while the coordinate vector of μ is $(q_1, q_2 - \tau_3)$. Their wedge determinant is therefore

$$(p_1 - \tau_3)(q_2 - \tau_3) - p_2 q_1.$$

□

5 Cycle 18 Component Split

Choose a basis $F = B + \alpha B$ with $\alpha^2 = \nu \in B$, and write

$$p_1 = p_{10} + \alpha p_{11}, \quad p_2 = p_{20} + \alpha p_{21}, \quad q_1 = q_{10} + \alpha q_{11}, \quad q_2 = q_{20} + \alpha q_{21}.$$

Split

$$\Delta = \Delta_0 + \alpha \Delta_1, \quad \Delta_0, \Delta_1 \in B[\tau_1, \tau_2, \tau_3].$$

Theorem 1 (Monic quadratic and linear alpha component). *In the setting above,*

$$\begin{aligned} \Delta_0 &= \tau_3^2 - (p_{10} + q_{20})\tau_3 + p_{10}q_{20} + \nu p_{11}q_{21} - p_{20}q_{10} - \nu p_{21}q_{11}, \\ \Delta_1 &= -(p_{11} + q_{21})\tau_3 + p_{10}q_{21} + p_{11}q_{20} - p_{20}q_{11} - p_{21}q_{10}. \end{aligned}$$

Consequently Δ_0 is monic quadratic in τ_3 , while Δ_1 has degree at most one in τ_3 .

Proof. Substitute the coordinate expressions for p_i, q_i into the identity of Lemma 1 and use $\alpha^2 = \nu$. Collecting the coefficient of 1 gives the displayed formula for Δ_0 , and collecting the coefficient of α gives the displayed formula for Δ_1 . The coefficient of τ_3^2 in Δ_0 is 1, and the α -component has no τ_3^2 -term. □

Corollary 1 (Graph reduction outside the zero-alpha branch). *If Δ_1 is not the zero polynomial, write*

$$\Delta_1 = s(\tau_1, \tau_2)\tau_3 + h(\tau_1, \tau_2).$$

On the open locus $s \neq 0$, the common-zero condition $\Delta_0 = \Delta_1 = 0$ is supported on the graph

$$\tau_3 = -h/s.$$

Proof. This is immediate from solving the linear equation $\Delta_1 = 0$ for τ_3 on $s \neq 0$. □

6 New Divisibility-Gate Theorem

The next theorem is the main new theorem extracted from the Cycle 18 reduction. It is stronger than merely saying that the branch is a graph: it shows that any two-dimensional family must satisfy an explicit divisibility identity.

Theorem 2 (Divisibility gate for the graph branch). *Assume the restricted setting above and assume $\Delta_1 \neq 0$. Write*

$$\Delta_1 = s(\tau_1, \tau_2)\tau_3 + h(\tau_1, \tau_2),$$

and define the denominator-cleared graph polynomial

$$G(\tau_1, \tau_2) = s(\tau_1, \tau_2)^2 \Delta_0(\tau_1, \tau_2, -h(\tau_1, \tau_2)/s(\tau_1, \tau_2)).$$

Then $\deg G \leq 4$. If G is not the zero polynomial, then

$$\#\{\tau \in B^3 : \Delta_0(\tau) = \Delta_1(\tau) = 0\} = O(p).$$

Consequently the corresponding number of distinct slopes satisfies

$$C_2 = O(p) = O(n).$$

Therefore a growing-prime family with

$$C_2 = \Theta(p^2) = \Theta(q_{\text{line}})$$

in the nonzero- Δ_1 branch can occur only if $G \equiv 0$.

Proof. Because the p_i, q_i are affine-linear in (τ_1, τ_2) , the coefficient s has degree at most 1, and h has degree at most 2. Since Δ_0 is monic quadratic in τ_3 , clearing the denominators after substituting $\tau_3 = -h/s$ gives a polynomial G of degree at most 4.

On $s \neq 0$, the equation $\Delta_1 = 0$ forces $\tau_3 = -h/s$. Hence every common zero on this open locus gives a base pair (τ_1, τ_2) with $G(\tau_1, \tau_2) = 0$. If G is nonzero, Schwartz–Zippel gives at most $4p$ such base pairs, and each base pair gives at most one value of τ_3 .

It remains to count the exceptional locus $s = 0$. There $\Delta_1 = 0$ also forces $h = 0$. If s is a nonzero affine form, this is contained in an affine line; if $s \equiv 0$, then h is a nonzero polynomial of degree at most 2. In either case there are $O(p)$ base pairs. For each such pair, Δ_0 is a monic quadratic in τ_3 , so it gives at most two values of τ_3 . The total common-zero count is $O(p)$.

Finally, the number of distinct slopes is bounded by the number of landing points τ . Thus $C_2 = O(p)$ whenever $G \not\equiv 0$, and any $\Theta(p^2)$ family in this branch must satisfy $G \equiv 0$. $\square$

Remark 1. *The identity $G \equiv 0$ says that, after clearing the s -denominator, Δ_0 is divisible by the linear equation Δ_1 on the graph branch. This is now the first exact algebraic gate that a large slope family must pass.*

7 Nearby Proven Gates

The same restricted lane contains two other useful gates. They are included here because they explain where the new theorem fits.

Proposition 1 (Base-component complete-intersection gate). *Suppose $\Delta_0, \Delta_1 \in B[\tau_1, \tau_2, \tau_3]$ have no common surface component. Then*

$$\#V(\Delta_0, \Delta_1)(B) = O(p),$$

and hence this branch contributes $C_2 = O(p)$ slopes.

Proof. The two components have bounded degree. If they share no surface component, their common zero set in $\mathbb{A}_B^3$ is a curve of bounded degree. A bounded-degree affine curve over $\mathbb{F}_p$ has $O(p)$ rational points, and slopes are bounded by landing points. $\square$

Proposition 2 (Cycle 16 determinant gate). *Let $Q(z_0, z_1)$ be the Cycle 15–16 determinant consistency polynomial for the affine system*

$$L_z(\tau) = \iota(\tau) - z\mu(\tau) = 0, \quad z = z_0 + \alpha z_1.$$

If Q is not identically zero, then in the restricted $D = \mathbb{F}_p$, $t = \sigma = 2$, $j = 3$, off- R_0 window,

$$C_2 \leq 4p = O(p) = O(n).$$

Proof. Cycle 16 gives $\deg Q \leq 4$ and shows that every landing slope must satisfy $Q(z_0, z_1) = 0$. If $Q \neq 0$, Schwartz–Zippel over $B = \mathbb{F}_p$ gives at most $4p$ possible pairs (z_0, z_1) , hence at most $4p$ slopes. $\square$

8 Slope Map and Remaining Heuristics

On the graph branch, the landing equation $\iota = z\mu$ gives

$$p_1 - \tau_3 = zq_1, \quad p_2 = z(q_2 - \tau_3).$$

Where $q_1 \neq 0$, substituting $\tau_3 = -h/s$ yields

$$z(\tau_1, \tau_2) = \frac{p_1 + h/s}{q_1}.$$

Equivalently, eliminating τ_3 recovers the Cycle 14 quadratic

$$q_1 z^2 - (p_1 - q_2)z - p_2 = 0,$$

so the residual map is controlled by

$$(\tau_1, \tau_2) \mapsto [q_1 : (p_1 - q_2) : p_2] \in \mathbb{P}^2(F).$$

Heuristic 1 (Exceptional identity should be rare). *For generic source data, the polynomial G should not vanish identically. Thus the graph branch should usually be curve-sized, and the first possible large-slope families should come from algebraically special data satisfying the divisibility identity $G \equiv 0$ or from the zero-alpha branch $\Delta_1 \equiv 0$.*

Heuristic 2 (What remains after the gate). *If $G \equiv 0$, the problem is no longer to show that the common-zero set is small. It may be surface-sized. The remaining question is whether the projective coefficient map has one-dimensional image on every source-valid surface, giving $C_2 = O(p)$, or whether some growing-prime source-valid family has two-dimensional image and $C_2 = \Theta(p^2)$.*

9 Next Experimental Checks

A useful scanner should compute, for each source-valid candidate:

$$\Delta_1 = s\tau_3 + h, \quad G = s^2\Delta_0(\tau_1, \tau_2, -h/s),$$

then record whether $G \equiv 0$, whether $\Delta_1 \equiv 0$, the exceptional locus $s = h = 0$, the image size of

$$[q_1 : (p_1 - q_2) : p_2],$$

and the direct slope count over distinct split triples $T \subset \mathbb{F}_p$.

The certificate should keep at least:

$$p, \ q_{\text{gen}}, \ q_{\text{line}}, \ E, \ B_{\text{num}}, \ \text{seed}, \ \text{stratum}, \ R_0\text{-status}, \\ \Delta_1 \equiv 0, \ G \equiv 0, \ \#\{s = h = 0\}, \ \#\text{image}, \ C_2, \ \#\{T \text{ split distinct}\}.$$

10 Non-Claims

This note does not claim:

- a proof of the corrected MCA conjecture;
- a positive theorem above corrected reserve;
- a q_{gen} local-limit theorem;
- a list, CA, MCA, line-decoding, SNARK, or protocol consequence;
- a large $\Theta(p^2)$ counterpacket.

The mathematical value is narrower: the possible large branch in this restricted window must satisfy an explicit algebraic identity before it can even be surface-sized.